

DAY 1 SUMMER TRAINING ASSIGNMENT

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1. What is a variable in C/C++?

A variable is a named memory location used to store data in a program. Its value can change while the program is running. Variables make it easy to store, update, and use information whenever required.

2. Why do we use variables?

Variables help us store values like marks, age, salary, and other information. They make programs flexible because the same code can work with different inputs. Without variables, every value would have to be written directly.

3. Difference between variable name and value?

A variable name is the identifier given by the programmer. The value is the actual data stored inside that variable. For example, age is the name and 20 can be its value.

4. Valid variable name?

Correct Answer: total_marks. It follows C++ naming rules because it starts with a letter and contains only letters and an underscore.

5. Why is Age different from age?

C++ is a case-sensitive language. This means uppercase and lowercase letters are treated differently. Therefore, Age and age are considered two different variables.

6. Can a variable contain spaces?

No. Spaces are not allowed in variable names because the compiler treats spaces as separators. We can use an underscore (_) instead.

7. Can a variable start with _?

Yes. A variable can begin with an underscore. However, meaningful names starting with letters are usually preferred for better readability.

8. Same variable name in same scope?

The compiler will show a redeclaration error. Two variables with the same name cannot exist in the same scope because it creates confusion.

9. Three meaningful variable names

studentName, rollNumber, studentMarks. These names clearly describe the purpose of each variable and improve code readability.

10. Better naming convention?

studentMarks is better because it clearly explains what the variable stores. Short names like sm are harder to understand in large programs.

Data Types

A data type tells the compiler what type of value a variable can store. Common data types are int, float, double, char, bool, and string. Choosing the correct data type improves memory usage and program performance.

Input/Output

cin is used to take input from the user using the >> operator. cout is used to display output using the << operator. endl moves the cursor to the next line after printing.

Operators

Operators perform different operations on data such as addition, subtraction, multiplication, comparison, and logical operations. Examples include +, -, *, /, %, ==, &&, || and !.

Program Answers (41–50)

41. Hello World

Prints 'Hello World' on the screen. It is the first program written by beginners to understand the basic structure of a C++ program.

42. Sum of Two Numbers

Take two numbers from the user using cin and display their sum using cout. This program demonstrates variables, input, output, and arithmetic operators.

43. Area of Rectangle

Formula: $\text{Area} = \text{Length} \times \text{Breadth}$. The program takes length and breadth as input and displays the calculated area.

44. Simple Interest Formula

Simple Interest = $(\text{Principal} \times \text{Rate} \times \text{Time}) / 100$. This formula is commonly used in banking and finance calculations.

45. Simple Interest Program

Take principal, rate, and time as input. Apply the SI formula and display the result. This program uses arithmetic operators and variables.

46. Temporary Variable

A temporary variable stores one value safely while swapping two numbers. Without it, one of the original values would be lost.

47. Swapping

Before swapping: $a = 10, b = 20$. After swapping: $a = 20, b = 10$.

48. Celsius to Fahrenheit

Formula: $F = (C \times 9/5) + 32$. This converts temperature from Celsius to Fahrenheit.

49. 0°C in Fahrenheit

The Fahrenheit value of 0°C is 32°F. This is obtained by applying the conversion formula.

50. Integer Division

Using $9/5$ performs integer division and removes the decimal part. Therefore, $9.0/5$ or 1.8 should be used for accurate results.

Bonus MCQs

Q1. c) int

Q2. b) float

Q3. b) 8

Q4. c) 1

Q5. b) &&